🍽️ Restaurant Recommendation System

Developed By: -	**Prn No.**
1.Aditya Dattatray Dange.	**2023011032017**
2.Shafika Shakil Nadaf.	**2023011032057**
3.Gauri Anil Chaougale.	**2023011032011**

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1. Introduction

Finding the right restaurant can be difficult because there are many options available. People often spend a lot of time searching for restaurants based on their taste, budget, and location.

The **Restaurant Recommendation System** is a smart web-based application that helps users discover restaurants based on their preferences, location, and ratings. The system uses a **Content-Based Filtering algorithm** to provide personalized restaurant suggestions.

This project is useful for:

- 👤 Restaurant Visitors
- 🍴 Restaurant Owners
- 🚚 Food Delivery Platforms

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2. Objectives of the Project

The main objectives of this project are:

- Implement a **Content-Based Recommendation System**
- Perform **data preprocessing and cleaning**
- Analyze the dataset using **data visualization**
- Apply **machine learning techniques**
- Evaluate model performance
- Build and deploy the system using the **Flask web framework**

3. System Overview

The system recommends restaurants by analyzing:

- Cuisine type
- User preferences
- Restaurant features
- Ratings and reviews

The goal is to provide **personalized suggestions** and improve the overall dining experience.

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4. Key Features

- 🔍 Personalized restaurant recommendations
- 📍 Location-based suggestions
- ⭐ Rating-based filtering
- 🧠 Content-Based Filtering algorithm
- 📊 Data visualization for insights
- 🌐 Web application using Flask
- 💡 Clean and user-friendly interface

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5. Technologies Used

Technology	Purpose
Python	Programming language
Flask	Web application framework
Pandas & NumPy	Data processing
Scikit-learn	Machine learning
Matplotlib / Seaborn	Data visualization
HTML, CSS,	Frontend development

JavaScript

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6. System Architecture

The workflow of the system:

User Input → Data Preprocessing → Feature Extraction → Similarity Calculation → Recommendation Output

Step Explanation

1. User enters preferences (cuisine, location, rating).
2. Data is cleaned and prepared.
3. Important features are extracted.
4. Similar restaurants are calculated.
5. Best matching restaurants are recommended.

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7. Project Structure

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Restaurant-Recommendation-System
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├── data/              Dataset files
├── static/            CSS, JavaScript, Images
├── templates/         HTML pages (Flask)
├── app.py             Main Flask application
├── model.py           Recommendation logic
├── requirements.txt   Dependencies
└── README.md          Project documentation
```

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8. Data Preprocessing

Data preprocessing is very important before building the model.

Steps performed:

- Handling missing values
- Removing duplicate records
- Feature selection

- Text processing (if required)

This improves the accuracy and quality of recommendations.

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9. Data Visualization

Visualization helps understand patterns in the dataset.

Charts created:

- Ratings distribution
- Popular cuisines
- Restaurant trends
- User preference analysis

These insights help in better decision making.

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10. Model Building

The recommendation model is built using **Content-Based Filtering**.

Feature Extraction

Two methods used:

- **TF-IDF Vectorizer**
- **Count Vectorizer**

Similarity Calculation

- **Cosine Similarity** is used to find similar restaurants.

Model Evaluation

- Accuracy metrics are used to evaluate performance.

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11. Web Application using Flask

The project is deployed as a web application.

How it works:

1. User enters preferences on the website.
2. Backend processes the request.
3. The system returns personalized restaurant recommendations.

This makes the system interactive and easy to use.

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12. How to Run the Project

Step 1: Clone Repository

git clone https://github.com/your-username/restaurant-recommendation-system.git
cd restaurant-recommendation-system

Step 2: Install Dependencies

pip install -r requirements.txt

Step 3: Run Application

python app.py

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13. Use Cases

👤 **For Users**

- Discover new restaurants
- Get personalized dining suggestions

🍴 **For Restaurant Owners**

- Increase restaurant visibility
- Target the right customers

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14. Advantages of the System

- Saves time in searching restaurants
- Provides personalized recommendations

- Improves user experience
- Helps restaurants reach the right audience

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15. Future Enhancements

- Add collaborative filtering
- Integrate real-time location services
- Add user login and reviews
- Mobile app development
- Integration with food delivery platforms

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16. Conclusion

The **Restaurant Recommendation System** successfully demonstrates how machine learning and web development can be combined to build a useful real-world application.

The system helps users discover restaurants easily and provides personalized recommendations based on their preferences. This project shows the practical use of **data science, machine learning, and web development** in solving everyday problems.